

Mining & Extraction: The Hidden Origins of AI Hardware

The minerals and metals that power artificial intelligence do not materialize from thin air. They are extracted from the earth through mining operations that span every continent, often in regions where environmental regulations are weak and labor protections are minimal. This sheet examines the key extraction sites behind AI's material footprint.

Cobalt — Democratic Republic of Congo

Mutanda Mine, Lualaba Province

The Mutanda mine, operated by Glencore until its temporary closure in 2019 and reopening in 2022, was one of the world's largest cobalt-producing sites. Located in the copper-cobalt belt of southern DRC, Mutanda produced approximately 27,300 tonnes of cobalt in 2018 alone, representing roughly 20% of global supply.

Artisanal Mining, Katanga Region

Alongside industrial operations, an estimated 150,000–200,000 artisanal miners (known as *creuseurs*) work in the Katanga region, many of them children. These miners dig by hand in unregulated tunnels, exposed to toxic dust containing cobalt, uranium, and other heavy metals.

Key Statistics

- The DRC supplies approximately 73% of global cobalt (USGS, 2023).
- An estimated 40,000 children work in DRC cobalt mines (Amnesty International, 2016).
- Artisanal and small-scale mining accounts for 15–30% of DRC cobalt output (BGR, 2019).
- Cobalt miners in the DRC earn on average \$1–2 per day (Amnesty International, 2016).
- A single EV battery requires 8–14 kg of cobalt; AI server batteries and electronics also depend on it.

Sources: Amnesty International, *"This Is What We Die For"* (2016); USGS Mineral Commodity Summaries (2023); German Federal Institute for Geosciences and Natural Resources (BGR), *Cobalt from the DRC* (2019).

Lithium — Chile, Salar de Atacama

The Salar de Atacama in Chile's Atacama Desert is the world's most productive lithium brine operation, operated primarily by SQM and Albemarle. Lithium is essential for the rechargeable batteries in servers, UPS systems, and portable devices that support AI infrastructure.

The extraction process pumps lithium-rich brine into massive evaporation ponds, consuming approximately 2,000 liters of water per kilogram of lithium produced. In a region where annual rainfall is below 15 mm, this has devastating consequences for indigenous Atacameño communities and local ecosystems.

Key Statistics

- Chile holds approximately 9.3 million tonnes of lithium reserves, the world's largest (USGS, 2023).
- The Salar de Atacama accounts for roughly 30% of global lithium production.
- Lithium extraction consumes an estimated 65% of the Salar de Atacama region's water (Babidge et al., 2019).

- Global lithium demand is projected to increase 5–6x by 2030 (IEA, 2023).

Sources: USGS Mineral Commodity Summaries (2023); IEA, *Global Critical Minerals Outlook* (2023); Babidge, S. et al., "Evaporation is theft", *Geoforum* (2019).

Rare Earth Elements — China, Bayan Obo

The Bayan Obo mine in Inner Mongolia is the world's largest rare earth mine, supplying elements such as neodymium, praseodymium, and dysprosium. These are critical for the permanent magnets in hard drives, speakers, and electric motors found throughout data center infrastructure.

Processing rare earths generates enormous volumes of toxic waste. The Bayan Obo tailings pond — a radioactive, chemically contaminated lake — spans approximately 11 km² and is visible from satellite imagery. Surrounding villages report elevated rates of cancer and respiratory illness.

Key Statistics

- China produces approximately 70% of global rare earth elements and processes roughly 90% (USGS, 2023).
- The Bayan Obo tailings pond contains an estimated 180 million tonnes of waste.
- Processing 1 tonne of rare earth oxide generates approximately 2,000 tonnes of toxic waste (China Society of Rare Earths).
- Thorium radioactive waste from Bayan Obo has contaminated the Yellow River basin.

Sources: USGS Mineral Commodity Summaries (2023); Huang, X. et al., "Rare earth pollution in Bayan Obo", *Journal of Environmental Radioactivity* (2016); China Society of Rare Earths.

Coltan — DRC, Rubaya

Rubaya, in North Kivu province, is the epicenter of coltan (columbite-tantalite) mining in the DRC. Tantalum refined from coltan is essential for capacitors in virtually every electronic device, including the GPUs and server boards that run AI workloads.

Mining in the Rubaya area is deeply entangled with armed conflict. Various militia groups have historically controlled mines and taxed miners, funding ongoing violence in eastern DRC. Despite certification schemes (iTSCI), traceability remains limited.

Key Statistics

- The DRC holds approximately 80% of known coltan reserves globally.
- An estimated 60% of coltan mining in eastern DRC has been linked to armed groups (UN Panel of Experts, 2020).
- Rubaya's mines produce an estimated 15–20% of global tantalum supply.
- Miners earn approximately \$2–5 per day in artisanal coltan operations.

Sources: UN Panel of Experts on the DRC (2020); IPIS, *Mapping Artisanal Mining Areas in Eastern DRC* (2019); Enough Project.

Nickel — Indonesia, IMIP Sulawesi

The Indonesia Morowali Industrial Park (IMIP) in Central Sulawesi is a massive nickel processing complex, largely Chinese-financed, producing nickel for stainless steel and increasingly for battery-grade nickel sulfate. Nickel is used in batteries, plating, and alloys in server hardware.

Indonesia's nickel boom has come at a severe environmental cost: deforestation of tropical rainforest, dumping of mine tailings into the sea (submarine tailings disposal), and dangerous working conditions. Furnace explosions at IMIP have killed multiple workers.

Key Statistics

- Indonesia produces approximately 49% of global nickel (USGS, 2023).
- Nickel mining in Sulawesi has caused the deforestation of over 25,000 hectares (Mighty Earth, 2023).
- At least 21 workers died in industrial accidents at IMIP between 2015 and 2023.
- Indonesia's nickel reserves total approximately 21 million tonnes, the world's largest (USGS, 2023).

Sources: USGS Mineral Commodity Summaries (2023); Mighty Earth, *Nickel Unearthed* (2023); Indonesian legal aid organizations (LBH).

Gold — Ghana (Galamsey) & Peru (Madre de Dios)

Gold is used in virtually all electronic connectors, circuit board traces, and bonding wires due to its excellent conductivity and corrosion resistance. Every GPU, CPU, and memory module contains small amounts of gold.

In **Ghana**, illegal artisanal gold mining (*galamsey*) has devastated rivers and cocoa farmland. Mercury amalgamation techniques poison waterways and miners alike. In **Peru's Madre de Dios** region, illegal gold mining in the Amazon has destroyed over 100,000 hectares of primary rainforest, creating mercury contamination visible from space.

Key Statistics

- A single smartphone contains approximately 0.03 g of gold; a server significantly more.
- Galamsey has polluted an estimated 60% of Ghana's water bodies (Ghana Water Resources Commission, 2021).
- Over 100,000 hectares of Amazon rainforest destroyed in Madre de Dios (CINCIA, 2021).
- Artisanal gold mining releases approximately 1,000 tonnes of mercury annually worldwide (UNEP, 2018).

Sources: UNEP, *Global Mercury Assessment* (2018); CINCIA (Centro de Innovación Científica Amazónica), 2021; Ghana Water Resources Commission.

Copper — Chile, Escondida Mine

The Escondida mine in Chile's Atacama Desert, operated by BHP, is the world's largest copper mine. Copper is fundamental to all electronics — wiring, circuit boards, heat sinks, and power distribution in data centers.

Escondida consumes vast quantities of water and energy, and has faced repeated labor disputes. The mine has increasingly turned to desalinated seawater pumped 170 km uphill at enormous energy cost.

Key Statistics

- Escondida produced approximately 1.08 million tonnes of copper in 2022, roughly 5% of global supply (BHP, 2023).
- A typical data center uses 20–40 tonnes of copper (Copper Development Association).
- Chile produces approximately 24% of the world's copper (USGS, 2023).
- Escondida's desalination plant consumes approximately 3,200 liters of diesel per hour.

Sources: USGS Mineral Commodity Summaries (2023); BHP Annual Report (2023); Copper Development Association.

Polysilicon — Xinjiang, China

Approximately 35% of global polysilicon — the ultra-pure silicon used in semiconductor wafers — is produced in the Xinjiang Uyghur Autonomous Region of China. Companies such as Daqo New Energy, GCL-Poly, and East Hope Group operate massive polysilicon plants in the region, benefiting from cheap coal-fired electricity.

These operations have been linked to forced labor of Uyghur and other Turkic minorities through government-organized labor transfer programs. In 2021, the United States passed the Uyghur Forced Labor Prevention Act, imposing restrictions on goods from Xinjiang.

Key Statistics

- Xinjiang produces approximately 35% of global polysilicon (Sheffield Hallam University, 2021).
- Polysilicon production is extremely energy-intensive: ~100 MWh per tonne (Bernreuter Research).
- Xinjiang's electricity mix is approximately 65% coal-fired.
- Over 1 million Uyghurs have been detained in internment facilities (UN OHCHR, 2022).

Sources: Murphy, L. & Elimä, N., *"In Broad Daylight"*, Sheffield Hallam University (2021); UN OHCHR Assessment of Xinjiang (2022); Bernreuter Research; Daqo New Energy SEC filings.

Conclusion

The extraction stage reveals a pattern: the materials that enable AI are sourced disproportionately from the Global South, in conditions involving environmental destruction, health hazards, child labor, armed conflict, and forced labor. Understanding these origins is essential for any serious assessment of AI's true cost.